Balanced districting for a national sales channel: the model, the two-stage scheme, and the certified draw

Territory-division project*

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Abstract

The business is standing up a *new* "national" channel — the two largest product manufacturers carved out of the financial-institutions and wirehouse channels — and wants it cut into territories of roughly equal market opportunity. This note develops the mathematics of that problem and reports the first certified draw. We generalise the utility model to n representatives per postal code and show it reduces identically to the two-player model the programme was built for; we prove the model is invariant under a global rescaling of the dollar fields, which licenses exporting a descaled instance from the confidential machine; we prove the central structural fact, that *maximum Nash welfare on a common measure is exactly equal-size districting*, so the $\sim$\$1B target is an emergent property rather than a constraint; and we decompose total welfare into a partition-invariant part and an incumbency premium. The real export (2026-09-01) then rewrote the instance: 1,229 zips, 111 representatives, $\approx$\$13B of opportunity, $k = 13$ — and a *national* footprint whose sold-zip adjacency graph has 547 components, which makes a contiguity constraint vacuous rather than binding. Stage 1 is therefore *center-based balanced assignment* on planar equal-area coordinates. We prove the transportation-relaxation lemma that makes it work — with the centers fixed, the relaxed assignment LP is a transportation problem, so a basic optimum splits at most $k - 1$ zips — and give three post-hoc certificates for a drawn map, with the real run's numbers: the draw sits 4.510×10^{-5} nats below an analytic balance ceiling that bounds every partition. The superseded contiguity formulation and the component-allocation ceiling are retained, demoted, at the end.

1 The instance and the re-scoping

The programme's original problem was bilateral. Two sales forces merge; a national *census* intersects the two legacy representative maps and decomposes the country into contested components, each with one A-rep and one B-rep; inside a component the zips are divided between them by maximum Nash welfare subject to hard contiguity. Everything downstream — one binary per zip, a scalar exchange rate $u_a(z)/u_b(z)$, the separator cut with its "a branch" and its "b branch" — inherits that two-sidedness.

*A self-contained companion to `docs/CHANNEL.md` (the problem), `docs/MODEL.md` (the model) and `docs/DATA.md` (the descaled export route), which are the source of record. The implementations are `td/model.py` (the n-way primitives), `td/channel.py` (stage 2 and the ceiling), `td/solvers/centers.py` (stage 1) and `td/solvers/cert_draw.py` (the certificates), with `tests/test_model.py`, `test_channel.py`, `test_centers.py` and `test_cert_draw.py` asserting the identities proved below. Every derived number in this note is computed by `ceiling.py` and injected as a macro; the handful that only a solver run can produce (node counts, wall-clock times) are frozen there as literals, cited to their run, and cross-checked against that run's `metrics.json` when it is on disk.

The new problem is not a merger. A *national channel* is being created by carving the two largest manufacturers out of the existing financial-institutions and wirehouse channels, and its footprint is to be divided into territories of roughly equal opportunity. Two structural consequences follow. First, *there is no pair structure to decompose along*: the census's components came from the overlap of two legacy maps, and a greenfield channel has no such overlap, so what was many independent 30–500 zip subproblems becomes one k-way partition of the whole footprint. Second, *the objective's argument changes measure*: what is to be balanced is *opportunity*, a quantity attached to the territory and not to any representative — the same for every candidate owner. That single change turns the Nash criterion into a balance criterion exactly (Section 4), which is the main structural result of this note.

zips carrying sales	1,229 (1,223 with a gazetteer internal point)
distinct representatives	111
node classes	675 contested / 477 uncontested / 2 vacant / 75 untapped
total market opportunity	≈\$13B
footprint	**national**: west 33.2%, east 31.0%, TX 11.5%, FL 6.7%, rest 17.6%
sold-zip Rook adjacency	547 components; largest 5.1% of opportunity
territory target	$\sim$\$1B $\Rightarrow k = 13$ at \$1.00B each

What the export changed. The instance above is the descaled real export of 2026-09-01. It corrected the 2026-08-31 sizing on three points, and the third is the one that changed the algorithm. *(i) Size.* The earlier count of 2,232 sold zips was a double-count in the source pull; the real figure is 1,229. *(ii) Scale and headcount.* Total opportunity is ≈\$13B, not \$6.2B, and there are 111 representatives, not 72; at a \$1B target that moves k from 6 to 13. *(iii) Geography.* The footprint is *national*. The premise that the channel covered the two coasts, Texas and Florida with the midwest uncovered was empirically false: 17.6% of opportunity sits outside those four regions. There are no four islands, and Section 9's component-allocation planning ceiling has nothing to allocate over.

Three readings of the table matter.

First, *scale sits at the frontier*: 1,229 units against 13 districts is ≈15,977 assignment binaries. The unit count is of the same order as the $\sim$1,500 units Validi and Buchanan (2022) certify for political districting with lazily separated connectivity cuts, and an order of magnitude above the 135 zips the programme's own single-tree solver certifies on the two-player problem — but at $k = 13$ districts rather than two, which is where the binaries come from.

Second — and this is the decisive one — *adjacency contiguity is not a usable constraint on this footprint*. The channel sells in 1,229 zips scattered across the country, so the Rook adjacency graph *restricted to sold zips* has 547 components: the largest holds 5.1% of opportunity and 68% of opportunity sits in components holding under 1% each. A contiguous k-partition of that graph does not exist for any k below several hundred, and at $k = 13$ the constraint is not binding-but-hard, it is *infeasible*. Restoring contiguity would mean re-exporting the unsold "glue" zips — most of the country — and districting an object the channel does not sell in. The decision (2026-09-01) is therefore to drop adjacency and keep the geometric requirement that contiguity was standing in for: **compactness**. Stage 1 becomes center-based balanced assignment on planar equal-area coordinates (Section 7); the contiguity formulation is retained, demoted, in Section 10.

Third, granularity is benign. The largest single zip carries 1.07% of total opportunity, about 14% of one $k = 13$ district, so near-perfect balance is not obstructed by indivisibility — a claim Section 8 turns into a certificate rather than leaving as an intuition.

A fourth reading is a business question rather than a mathematical one: 111 representatives against 13 territories is a 8.5:1 ratio. The reading that makes sense of it is that the channel is a *specialist carve-out* staffed by a handful of senior wholesalers while the remaining representatives keep covering the other manufacturers in the existing channel; the unmatched representatives of Section 6 are therefore *not selected for this channel*, not released. If that reading is wrong, stage 2's framing needs revisiting.

2 The utility model with n representatives

2.1 Definition

Fix a zip z. Let $S_i(z) \geq 0$ be representative i's booked production at z, let

$$T_z \;=\; \sum_j S_j(z) \tag{1}$$

be all production booked to a named representative there, let $S_{\text{free}}(z) \geq 0$ be production booked at z under no named representative (Section 2.3), and let $M_z > 0$ be the market opportunity. Two parameters translate these into value: the *transfer capture* $\theta \in [0,1]$, the fraction of another representative's book an inheriting representative retains, and the *headroom credit* $\lambda \in [0,1]$, the rate at which untapped opportunity counts against booked production. With the net headroom convention $c_1 = 1-\lambda$, $c_2 = \theta(1-\lambda)$,

$$\boxed{u_i(z) \;=\; c_1\, S_i(z) \;+\; c_2\big(T_z - S_i(z)\big) \;+\; c_{\text{free}}\, S_{\text{free}}(z) \;+\; \lambda M_z} \tag{2}$$

The inheriting representative keeps c_1 of their own book, captures c_2 of everyone else's, capitalises unowned book at a rate c_{free} decided separately, and is credited λ of the whole opportunity. Note that T_z in (1) runs over named representatives only; S_{free} enters (2) through its own term.[1]

Headroom generalises verbatim: the model requires

$$M_z \;\geq\; \max_i \big(S_i(z) + \theta\,(T_z - S_i(z))\big). \tag{3}$$

An *allocation* is a map $\text{own} : Z \to R$ from zips to representatives, and each representative's *gain* at disagreement point $d = 0$ is the bundle utility

$$g_i \;=\; \sum_{z:\,\text{own}(z)=i} u_i(z). \tag{4}$$

The criterion is maximum Nash welfare (MNW), the Nash bargaining solution (Nash, 1950) at $d = 0$, equivalently the Eisenberg–Gale objective (Eisenberg and Gale, 1959):

$$\max_{\text{own}} \sum_i \log g_i \qquad \text{subject to the geometric constraint.} \tag{5}$$

Nothing in the outer-approximation argument that makes (5) exactly solvable was ever two-player: log is concave, g_i is linear in the assignment, so this remains a convex MINLP with a genuine optimality certificate (Duran and Grossmann, 1986; Fletcher and Leyffer, 1994). The fairness guarantee survives the move for the same reason — Caragiannis et al. (2019) state Pareto efficiency and envy-freeness up to one good for n agents, not two.

[1] The implementation follows this exactly (`model.utilities`). Its headroom validator `model.headroom_violations` is deliberately *more* conservative: it folds S_{free} into T_z and treats the filler as a possible holder, so a zip whose orphaned book alone exceeds its opportunity is still flagged.

2.2 The model reduces to the two-player one

Proposition 1 (Two-representative reduction)**.** *Let $n = 2$ with $R = \{a, b\}$, $S_a(z) = A_z$, $S_b(z) = B_z$ and $S_{\text{free}} \equiv 0$. Then (2) is identically the two-player model,*

$$u_a(z) = c_1 A_z + c_2 B_z + \lambda M_z, \qquad u_b(z) = c_2 A_z + c_1 B_z + \lambda M_z,$$

and (3) is identically $M_z \geq \max(A_z + \theta B_z,\ B_z + \theta A_z)$.

Proof. $T_z = A_z + B_z$, so $T_z - S_a(z) = B_z$ and $T_z - S_b(z) = A_z$; substituting into (2) with $c_{\text{free}} S_{\text{free}} = 0$ gives the two displayed expressions. For (3), the maximum runs over $i \in \{a, b\}$ and gives $\max(A_z + \theta B_z,\ B_z + \theta A_z)$. □

This is not a formality. It is what keeps the entire committed corpus of two-player results — the paper's worked 50-zip instance, the C1–C9 battery, every certified harness row — interpretable under the new model, and it is asserted numerically rather than argued: `test_model.py::test_two_rep_reduction` requires every n-way primitive to agree with the frozen two-player `base.py` to floating-point equality on a two-representative instance.

2.3 Unowned book: the vacancy filler

Some territories currently have no assigned representative, and their production is booked under a *filler key*: a representative-shaped sentinel that carries real sales, real opportunity and a real manufacturer, but is not a person. It must never become a candidate owner. A filler key with an objective term would have the solver bargaining on behalf of an empty chair, and because $\sum_i \log g_i$ is unbounded below as any $g_i \to 0$, it would trade real representatives' welfare away to feed it.

The schema separates the two roles, which is what makes the exclusion cheap: T_z is computed from *all* books, but the assignment ranges over *candidates* only, and candidacy is $\text{cand}(z) = \{i : S_i(z) > 0\}$ with filler keys removed. Orphaned production therefore arrives as its own attribute S_{free} and is capitalised by whoever inherits the zip. Four node classes result, with the export's counts:

class	$\lvert\text{cand}(z)\rvert$	book	count
contested	≥ 2	real	675
uncontested	1	real	477
vacant	0	filler only	2
untapped	0	none	75

Contested zips are the decision problem; an uncontested zip's owner is forced under a legacy rule but it still carries utility; a vacant zip has real book and no incumbent, so nobody can claim it by legacy; an untapped zip is opportunity with no book at all. Under the two-stage scheme none of the four is dropped — stage 1 partitions on M_z alone, so every zip lands in a district by construction, and candidacy re-enters only at stage 2.

Choosing c_{free}. Three coefficients are defensible, and they give materially different maps, so the implementation takes the mode explicitly (`filler_capture`) rather than defaulting quietly. `theta` ($c_{\text{free}} = c_2$) discounts orphaned book exactly like a live representative's; conservative, and it assumes vacant business is as person-sticky as anyone else's. `opportunity` ($c_{\text{free}} = \lambda$) treats it as untapped market; defensible if the sales in vacant territories are house or inbound business with no relationship content. `full` ($c_{\text{free}} = c_1$) is *the recommended choice*: the reason $\theta < 1$ exists at all is that a *departing* representative pulls relationships away with them, and a vacancy has nobody left to pull — whatever book has survived an already-departed representative has, by definition, survived the departure, so discounting it again double-counts an attrition that has already happened. The current default is `theta`, only because it is the no-change case. It is probably not the right answer. At 2 vacant zips out of 1,229 the choice cannot move this instance's map; it is recorded because the model outlives the instance.

3 Scale invariance, and what it licenses

The real instance lives on a confidential machine and its dollar amounts cannot travel. The following proposition is why they do not have to.

Write the utility (2) in *share* form. With $s_i(z) = S_i(z)/M_z$, $t_z = T_z/M_z$ and $f_z = S_{\text{free}}(z)/M_z$,

$$u_i(z) \;=\; M_z\Big[\,c_1 s_i(z) + c_2\big(t_z - s_i(z)\big) + c_{\text{free}} f_z + \lambda\,\Big]. \tag{6}$$

Every dollar amount has been pushed into the single factor M_z.

Proposition 2 (Scale invariance of the model)**.** *Let $\mathcal{I} = \big(G, (S_i)_i, S_{\text{free}}, M\big)$ be an instance and, for $\kappa > 0$, let $\kappa\mathcal{I}$ scale every dollar field by κ (the graph, the candidate sets and $\theta, \lambda, c_{\text{free}}$ unchanged). Then for every allocation* own *and every $\rho \geq 0$:*

1. *$u_i^{\kappa\mathcal{I}}(z) = \kappa\, u_i^{\mathcal{I}}(z)$ and $g_i^{\kappa\mathcal{I}} = \kappa\, g_i^{\mathcal{I}}$;*
2. *$\sum_i \log g_i^{\kappa\mathcal{I}} = \sum_i \log g_i^{\mathcal{I}} + n\log\kappa$, an additive constant independent of* own*;*
3. *consequently* own *maximises $\sum_i \log g_i - \rho\, P(\text{own})$ on $\kappa\mathcal{I}$ if and only if it does on $\mathcal{I}$, and every objective* difference *— hence every optimality gap and every certificate width in nats — is identical on the two instances;*
4. *the headroom condition (3) holds on $\mathcal{I}$ iff it holds on $\kappa\mathcal{I}$.*

Proof. (1) Each of the four terms of (2) is homogeneous of degree one in the dollar fields, so u_i is; g_i is a sum of such terms over a fixed set, hence also. (2) Take logarithms: $\log(\kappa g_i) = \log g_i + \log\kappa$, and sum over the n representatives. (3) The penalty $P(\text{own})$ — a perimeter count, or the moment of inertia of Section 7 once its own weights are descaled — does not involve the dollar fields, so the full objective shifts by the constant $n\log\kappa$; adding a constant changes neither the argmax nor any difference of objective values. (4) Both sides of (3) are homogeneous of degree one. $\square$

This is what makes a *descaled export* exact rather than approximate. The exporter emits shares $s_i(z) \in [0, 1]$ and a relative opportunity $m_{\text{rel}}(z) = M_z/\kappa$, κ being the median positive opportunity, and refuses to write a currency amount or the divisor. The loader reconstructs $S_i = s_i \cdot m_{\text{rel}}$ and $M = m_{\text{rel}}$, i.e. exactly $\mathcal{I}/\kappa$ up to six-significant-figure rounding, so by Proposition 2 it has the *same* optimal allocations, gaps and certificates as the real instance. Removing the scale drops a constant the solver never reads. That is strictly better than the synthetic twin it supersedes: real ZCTAs, real coordinates, real territory structure, real share patterns, and no rank-jitter or fidelity-audit apparatus at all. Every district mass quoted in Section 8.4 is in descaled units for this reason; only ratios, spreads and nats are reported, and all three are scale-free.

Remark 1 (What *is* scale-dependent)**.** Part (3) of Proposition 2 holds for every $\rho \geq 0$, including $\rho > 0$. This corrects a claim recorded earlier in the programme, that a positive compactness weight "mixes a log-scale term with a raw perimeter count and is therefore scale-dependent". For the Nash objective it is not: the log-sum's response to a rescale is a constant, which cancels in every comparison. Three genuinely scale-dependent things remain, and they are the ones to carry forward. *(i) The other districting entry point:* `districting.solve_contiguous` minimises a Kalai–Smorodinsky gap *plus* ρ·perimeter with the gap in raw gain units (Kalai and Smorodinsky, 1975); there κ multiplies one term and not the other, so that objective's argmax really does move. *(ii) ρ is a calibration constant, not a model parameter:* the battery's $\rho = 2 \times 10^{-3}$ knee prices one boundary edge against a fixed number of nats on a particular family of synthetic instances, and nats per edge depends on n and on the value distribution, neither of which survives the move to this footprint. Any ρ carried over must be re-derived. *(iii) Solver tolerances are absolute:* a feasibility tolerance of 10^{-9} is stated in gain units and gains scale with κ, whereas the certificate tolerance of 10^{-8} nats does not. This is why both stage-1 programmes descale their mass rows and their cost coefficients before calling HiGHS: scaling a constraint row together with its right-hand side leaves the feasible set identical, and scaling the objective by a positive constant leaves the argmin identical, so only the arithmetic changes.

One further caveat: the heavy-tail calibration of the synthetic generator (double Pareto-lognormal noise on sales *values*) does not transport either. Shares are bounded in $[0, 1]$ with quite different tail behaviour, so those knobs need recalibrating, not re-pointing.

4 Nash welfare on a common measure is equal-size districting

This is the centrepiece. In the two-player problem the objective and the balance of the two territories were different things; the Nash criterion was chosen for its axioms and its efficiency, and balance was a by-product. On a *common measure* they coincide exactly.

Let Z be the footprint with $M_z > 0$, and let $\mathcal{A} = (A_1, \ldots, A_k)$ be a partition of Z into k districts, with masses

$$M_j \;=\; \sum_{z \in A_j} M_z.$$

Proposition 3 (MNW on a common measure equalises)**.** *For every partition of Z into k parts, $\sum_j M_j = M(Z) := \sum_{z \in Z} M_z$ — the total is* partition-invariant*. Consequently*

$$\sum_{j=1}^{k} \log M_j \;\leq\; k\, \log \frac{M(Z)}{k}, \tag{7}$$

with equality if and only if $M_1 = \cdots = M_k = M(Z)/k$. Hence a partition attaining equal masses, when one exists, is a maximiser of the Nash objective, and every maximiser is as close to equal as the feasible set permits.

Proof. Partition-invariance is immediate: each z lies in exactly one part, so

$$\textstyle\sum_j M_j = \sum_j \sum_{z \in A_j} M_z = \sum_{z \in Z} M_z.$$

For the inequality, relax to $m \in \mathbb{R}^k_{>0}$ with $\sum_j m_j = M(Z)$ and maximise $\sum_j \log m_j$. The objective is strictly concave and the constraint set is compact and convex, so the maximiser is unique. The Lagrangian $L(m,\mu) = \sum_j \log m_j - \mu\big(\sum_j m_j - M(Z)\big)$ is stationary when $1/m_j = \mu$ for every j, so $m_j = 1/\mu$ for all j, and the constraint forces $m_j = M(Z)/k$. Substituting gives (7). Equivalently and without calculus, AM–GM gives $\big(\prod_j m_j\big)^{1/k} \le \frac{1}{k}\sum_j m_j = M(Z)/k$ with equality iff all m_j are equal; take logarithms and multiply by k. Since every partition's mass vector is feasible for the relaxation, the bound applies to it, and the equality case identifies exactly the equal partitions. $\square$

So the Nash objective *is* the balance objective on a common measure. It is not an approximation of it, and it is not a proxy: the argmax is the same. That is the whole justification for treating \$1B as an emergent target rather than a hard band — set $k = \lceil \text{total opportunity}/\$1\text{B} \rceil = 13$ and balance falls out of the criterion the project already uses for fairness.

Two refinements make the statement useful when perfect equality is unattainable, which on a real instance it always is.

Corollary 4 (Strict Schur-concavity: the objective always prefers the flatter vector)**.** *$\Phi(m) = \sum_j \log m_j$ is strictly Schur-concave on $\mathbb{R}^k_{>0}$. Hence if the mass vector m of one partition* majorises *the mass vector m' of another — m is the less even of the two, in the standard partial order — then $\Phi(m) \le \Phi(m')$, strictly unless m is a permutation of m'.*

Proof. Φ is symmetric and strictly concave, being a sum of a strictly concave function applied coordinatewise. A symmetric concave function is Schur-concave: for $i \ne j$ the Schur–Ostrowski condition $(m_i - m_j)\big(\partial_i\Phi - \partial_j\Phi\big) = (m_i - m_j)\big(1/m_i - 1/m_j\big) = -(m_i - m_j)^2/(m_i m_j) \le 0$ holds, with equality only at $m_i = m_j$; strictness of the inequality off the diagonal gives strict Schur-concavity. $\square$

Corollary 4 is what makes (5) a usable objective on a constrained feasible set: among admissible partitions the criterion orders any two comparable mass vectors by evenness, so the solver's search is a search for balance even when the perfectly balanced vector is not attainable.

Observation 1 (Verified by enumeration)**.** *`test_channel.py`'s `test_nash_welfare_is_equal_size_districting` brute-forces every way of cutting a 12-vertex path with uniform opportunity per zip into three* contiguous *blocks and checks that the argmax of $\sum_j \log M_j$ is the balanced cut $(4, 8)$, with reported relative spread exactly zero. It is a small check, but it is the one that would fail first if the identity were ever broken by a change to the objective.*

Why not just minimise a spread? Because minimising a dispersion measure is not an efficiency criterion and can leave *everybody* worse off — the project's trap 2. Re-verified at $d = 0$ on the paper's 50-zip instance: an unconstrained equalising MILP over all subsets reaches a Kalai–Smorodinsky gap of 5.2×10^{-8} at 98.1% of attainable welfare with gains (7.3226, 7.0699), and is *Pareto-dominated* by Nash's (7.3715, 7.2318) — both parties strictly prefer the Nash allocation to the "perfectly fair" one. Proposition 3 gives the balance without the pathology, because a strictly concave increasing objective stays on the Pareto frontier by construction. A hard band \$1B $\pm\, \epsilon$ would import the pathology back as an infeasibility risk.

5 The welfare decomposition

Proposition 3 concerns a common measure. Real utilities (2) are not common — they differ across representatives through the book terms. The following decomposition says exactly how much that matters.

Proposition 5 (Welfare decomposition)**.** *For any allocation* own *of all of* Z*,*

$$\sum_i g_i \;=\; \underbrace{\sum_{z\in Z}\Big[\lambda M_z + c_2 T_z + c_{\mathrm{free}} S_{\mathrm{free}}(z)\Big]}_{W_0,\ \textit{partition-invariant}} \;+\; \underbrace{(c_1-c_2)\sum_{z\in Z} S_{\mathrm{own}(z)}(z)}_{\textit{incumbency premium}}\,. \tag{8}$$

The first term does not depend on own *at all. The second, since* $c_1 - c_2 = (1-\theta)(1-\lambda) \geq 0$*, is maximised by giving every zip to the candidate holding the most book there.*

Proof. By (4), $\sum_i g_i = \sum_i \sum_{z:\mathrm{own}(z)=i} u_i(z) = \sum_{z\in Z} u_{\mathrm{own}(z)}(z)$, since each z contributes to exactly one representative. Expanding (2) at $i = \mathrm{own}(z)$ and regrouping,

$$u_{\mathrm{own}(z)}(z) = c_2 T_z + c_{\mathrm{free}} S_{\mathrm{free}}(z) + \lambda M_z + (c_1 - c_2) S_{\mathrm{own}(z)}(z),$$

in which only the last term mentions own. Summing over z gives (8). For the maximisation, $c_1 - c_2 = (1-\lambda) - \theta(1-\lambda) = (1-\theta)(1-\lambda) \geq 0$, so the sum is maximised termwise by $\mathrm{own}(z) \in \arg\max_i S_i(z)$ over the candidates. □

Read plainly: **the objective is "balance the territories", plus "where there is slack, leave business with the representative who already has it".** That is a reassuringly operational reading of a Nash bargaining criterion, and it is exact.

5.1 Sizing the two terms

The relative weight of the two halves of (8) decides how much the legacy books can move the map at all, and it is set by *saturation* $t_z = T_z/M_z$. At the reference parameters $\theta = 0.40$, $\lambda = 0.30$ — so $c_1 = 0.70$, $c_2 = 0.28$, $c_1 - c_2 = 0.42$ — and a saturation of ∼29.6%, a zip whose whole book sits with one incumbent is worth

$$u_{\text{incumbent}} = 0.507\,M_z, \qquad u_{\text{other candidate}} = 0.383\,M_z.$$

Of the incumbent's utility, 59.1% is the pure opportunity term λM_z and only 24.5% is the incumbency premium $(c_1 - c_2)S_i(z)$; the utility swing between holding the book and not holding it is 32.5%.

The consequence for the algorithm is decisive. Combining Propositions 3 and 5 through the AM–GM inequality of (7),

$$\sum_i \log g_i \;=\; n\log\!\Big(\frac{W_0 + (c_1-c_2)\sum_z S_{\mathrm{own}(z)}(z)}{n}\Big) \;-\; \underbrace{\Big[n\log\bar g - \sum_i \log g_i\Big]}_{D(g)\;\geq\;0}, \tag{9}$$

where $\bar g$ is the arithmetic mean of the gains and $D(g) \geq 0$, the log of the arithmetic-to-geometric mean ratio, vanishes exactly at perfect balance. The first term is bounded within a 24.5%-scale window by the incumbency premium, and on the live $k = 18$ draw that window is not small: to first order the premium ladder's map gap (this staff's shortfall from its best achievable map) costs 0.663 nats and its roster gap (this staff's further shortfall from the best-staff ceiling) another 0.249 nats, and evaluating the first term of (9) exactly at the two ends of that ladder puts the combined window at 0.890 nats. On the same draw $D(g) = 0.148$ nats. The 10^{-4}–10^{-2} range one might expect belongs to the *mass* imbalance $D(M)$, which is 1.5×10^{-4} nats at a mass spread of 1.4%; (9) is written on g, not on M, and the realised *gain* spread on this draw is ~60%. On the region the programme actually operates in the ordering therefore *inverts* — by a factor of 6.0, not by orders of magnitude: incumbency, not balance, is the larger term. The two-stage scheme survives as a business constraint — *territories shall be opportunity-balanced* — and not, as this section previously claimed, as a consequence derived from (9).

This ratio must be checked against real saturation. The live instance measures 29.6%, which sits just *below* the 30% threshold this note set for a "modest continuity tilt": saturation alone does not settle the question either way. What settles it is the directly measured premium window above — 0.890 nats against $D(g) = 0.148$ — which says incumbency book carries real weight in the objective whatever side of the 30% line the instance happens to fall on.

6 The two-stage scheme

stage	problem	status
1 — draw	k balanced compact districts on opportunity alone	heuristic, certified
2 — match	assign representatives to districts	exact

6.1 Stage 2 is a linear assignment problem on logs

Given a drawn map $A_1, \ldots, A_k$, let

$$g_{ij} \;=\; \sum_{z \in A_j} u_i(z) \tag{10}$$

be what district j is worth to representative i — evaluated for *every* representative on *every* district, unrestricted by legacy candidacy, which is the entire point of drawing the map before staffing it.

Proposition 6 (Nash-optimal staffing is a linear assignment problem)**.** *Suppose $g_{ij} > 0$ for all i, j. Maximising $\sum_i \log g_{i\sigma(i)}$ over injections σ from representatives to districts is a maximum-weight bipartite matching with weights $w_{ij} = \log g_{ij}$, and is therefore solved exactly in $O(\max(m,k)^3)$ by the Hungarian algorithm, where m is the number of representatives. When $m > k$ the matching is rectangular, and the unmatched representatives are exactly those not staffing the channel.*

Proof. The objective is additively separable over the pairs $(i, \sigma(i))$ that σ selects, the coefficient of (i, j) being $\log g_{ij}$, independent of the rest of σ. The injections σ are exactly the matchings of the complete bipartite graph $K_{m,k}$ saturating its smaller side. Hence this is the linear assignment problem with weight matrix $(\log g_{ij})$, which the Hungarian algorithm solves to optimality in cubic time; the rectangular case is the standard padding reduction. Positivity of g_{ij} is needed for $\log g_{ij}$ to be finite, and is checked rather than assumed. $\square$

Note that this is genuinely a different objective from utilitarian matching, and the difference is the reason to insist on it.

Example 1 (Nash and utilitarian matching disagree)**.** Two representatives, two districts, $g = \begin{pmatrix} 100 & 10 \\ 90 & 1 \end{pmatrix}$. The identity matching has utilitarian value 101 and Nash value 4.605; the swap has utilitarian value 100 and Nash value 6.802. The utilitarian rule picks the identity, handing representative 2 a district worth 1 to them because the *total* looks marginally better. The Nash rule picks the swap. A staffing that hands a senior wholesaler a territory containing almost none of their book is exactly the failure mode the channel cannot afford.

6.2 The cost of splitting, and the portfolio mitigation

Stage 1 cannot see relationships, so a good staffing may simply not be available on the map it draws. This is the same objection the programme raises to "decouple fairness from compactness": a two-stage scheme *relocates* the difficulty rather than removing it, and the joint optimum over (map, staffing) is in general strictly better than the sequential one — it is attained by first-stage maps that a balance-only stage 1 has no reason to prefer among its own optima.

The mitigation is cheap precisely because stage 2 is milliseconds: generate a *portfolio* of stage-1 draws — here one per random seed, elsewhere the incumbent sequence of a branch-and-cut solver or a recombination-style sampler (DeFord et al., 2021) — score each by its best staffing value, and keep the best pair. This is not the joint optimum and should not be reported as one; it is a cheap lower bound on it, and the spread between the best and worst draw in the portfolio diagnoses how much the split is costing. Section 8.4 reports a run in which the portfolio changed the answer: the draw that staffed best was *not* the stage-1-best draw.

A pleasant side effect: stage 1 needs almost none of the confidential data — only (z, M_z) and public coordinates, with opportunity plausibly third-party market sizing — while stage 2 needs the books but not the geometry, and can therefore run entirely on the work machine given only the returned district map.

7 Stage 1 as center-based balanced assignment

Contiguity is unavailable on this footprint (Section 1), and it was never the requirement — it was a proxy for one. What a sales territory actually needs is that a representative can cover it: a tight cluster of zips around a place, not a connected subgraph of a lattice. So stage 1 optimises *compactness* directly, in the centre-based formulation Hess and Samuels (1971) introduced for exactly this problem and the districting literature has used since (Zoltners and Sinha, 2005; Ríos-Mercado and Fernández, 2009; Duque et al., 2011).

7.1 The programme

Let $x_z \in \mathbb{R}^2$ be zip z's internal point in an equal-area planar projection, so that squared Euclidean distance is a real area-weighted moment and not a lat/lon approximation. With k

centers $c_1, \ldots, c_k \in \mathbb{R}^2$ and binaries $y_{zj} = 1$ iff zip z joins district j:

$$\begin{aligned}
\min_{y} \quad & \sum_{z \in Z} \sum_{j=1}^{k} M_z \, \|x_z - c_j\|^2 \, y_{zj} \\
\text{s.t.} \quad & \sum_{j=1}^{k} y_{zj} = 1 && \forall z \in Z, \\
& \sum_{z \in Z} M_z \, y_{zj} \;=\; \frac{M(Z)}{k} && \forall j \in \{1, \ldots, k\}, \\
& y_{zj} \in \{0, 1\}.
\end{aligned} \tag{11}$$

The mass equalities are *hard*, so any feasible point is exactly balanced and the objective only chooses among balanced maps; in the band form used in practice they are relaxed to $|\sum_z M_z y_{zj} - M(Z)/k| \le \delta$. By Proposition 3 this is the right division of labour: balance *is* the Nash objective on a common measure, and compactness is the tie-break among its optima — exactly the ordering (9) derives. Centers are then updated Lloyd-style, each to the M-weighted centroid of its district, and (11) re-solved; the loop stops when the labels repeat.

Two things are bought by the reformulation. The $k!$ label symmetry that would cripple branch-and-bound on an anonymous-district model is gone — district j is the one at c_j, so the labels are pinned by geography, which is why Hess's formulation is the standard remedy for it. And the constraint matrix of the relaxation is a network matrix, which is the content of the next lemma.

7.2 The transportation-relaxation lemma

Lemma 7 (Fixed centers: the relaxation is a transportation problem)**.** *Fix the centers and relax $y_{zj} \in \{0, 1\}$ to $y_{zj} \ge 0$ in (11). Then:*

1. *the substitution $\eta_{zj} = M_z \, y_{zj}$ turns the constraints into a Hitchcock transportation problem with supplies M_z, demands $M(Z)/k$ and no upper bounds — the bounds $y_{zj} \le 1$ are implied;*
2. *the relaxation is feasible, and every basic solution has acyclic support, hence at most $n+k-1$ positive entries, where $n = |Z|$;*
3. *consequently at most $k-1$ zips are* split*, i.e. have $y_{zj} > 0$ for two or more j. Every other zip is assigned integrally by the relaxation itself.*

Proof. (1) Substituting $\eta_{zj} = M_z y_{zj}$ ($M_z > 0$) sends $\sum_j y_{zj} = 1$ to $\sum_j \eta_{zj} = M_z$ and $\sum_z M_z y_{zj} = M(Z)/k$ to $\sum_z \eta_{zj} = M(Z)/k$; supplies and demands both total $M(Z)$, so the problem is a balanced transportation problem, and $\eta_{zj} \ge 0$ with $\sum_j \eta_{zj} = M_z$ forces $\eta_{zj} \le M_z$, i.e. $y_{zj} \le 1$. (2) $\eta_{zj} = M_z/k$ is feasible, so the feasible set is nonempty; it is bounded, so a basic optimal solution exists. The constraint matrix is the node–arc incidence matrix of the complete bipartite network on the n supply nodes and k demand nodes, and a set of its columns is linearly independent iff the corresponding edge set contains no cycle; hence the support of a basic solution is a forest on $n+k$ nodes and has at most $n+k-1$ edges. (3) Let F be the number of split zips. Every zip node has degree at least one in the support, since $M_z > 0$ must be shipped somewhere, and a split zip has degree at least two. Counting edges, $n + F \le \#\text{edges} \le n + k - 1$, so $F \le k - 1$. $\square$

Corollary 8 (Rounding damage is confined to $k-1$ zips)**.** *Rounding a basic optimum by giving each split zip to the district holding its largest share changes at most $k-1$ of the n assignments; every other zip keeps the district the exactly balanced relaxation gave it, and no district's mass moves by more than the total mass of the split zips, at most $(k-1)\max_z M_z$.*

Proof. Immediate from Lemma 7(3): the rounded solution differs from the basic optimum only on split zips, and the basic optimum's masses are exactly $M(Z)/k$, so each district's mass changes by at most the mass it gains or loses from those $F \le k-1$ zips. □

The crude bound $(k-1)\max_z M_z$ is not the operative fact — $\max_z M_z$ is 1.07% of $M(Z)$ here, so the bound is weak. What matters is the *count*: at $k = 13$ over 1,223 zips, rounding touches at most 12 zips, under 1% of the map, and the real run hit that bound exactly (12 split zips in the winning draw's final LP). A greedy repair pass then recovers what the rounding cost. That is why a heuristic can land within 4.510×10^{-5} nats of a bound that holds for every partition (Section 8.4): the only unbalanced part of the answer is $k-1$ zips wide.

7.3 The rest of the pipeline

Three components surround (11), and each is a heuristic that Section 8 then measures rather than trusts.

Seeding. M-weighted k-means++: the first center is a zip drawn with probability proportional to M_z, each further center with probability proportional to $M_z\, d^2(z, \text{nearest chosen center})$. Weighting by M rather than by zip count is the right prior when districts are equal in *mass*: a dense but light region should not attract centers the way its zip count would suggest.

The Lloyd loop. Assign by (11), move each center to its district's M-weighted centroid, repeat. Unlike plain k-means there is no drift to a lopsided fixed point, because every round's assignment is exactly balanced before rounding; the rounds buy compactness only.

The polish. A greedy pass over single-zip moves that strictly increase $\sum_j \log M_j$, restricted to destinations among the 3 nearest centers — the compactness guard, without which a Nash-greedy move would send a California zip to a Florida district for an epsilon of balance. Ties in the objective are broken by the more compact destination, and a district is never emptied (that would send the objective to $-\infty$).

Zips with no coordinates. Six of the 1,229 zips carry no gazetteer internal point (retired or non-ZCTA codes) and so are absent from the geometry. They are placed afterwards by a state-plurality rule: each goes to the district holding the plurality of already-placed zips from its own state, with ties and unknown states broken toward the district of smallest total mass, one zip at a time with counts and masses updated after each placement.

8 Certifying a draw

Nothing in Section 7 proves anything: the seeding is random, the rounding of the $k-1$ split zips is arbitrary, and the polish is a local search. Three post-hoc certificates say how far from optimal a drawn map actually is — and, as importantly, which questions they leave open. They are implemented in `td/solvers/cert_draw.py` and exercised against brute force at small k in `tests/test_cert_draw.py`.

8.1 (i) The balance ceiling: analytic, and valid for every partition

Proposition 9 (Balance ceiling at fixed k)**.** *Let Z be any finite set with $M_z > 0$ and let $A_1, \ldots, A_k$ be any partition of it into k parts of positive mass. Then*

$$\sum_{j=1}^{k} \log M_j \;\leq\; k \log \frac{M(Z)}{k},$$

with equality iff every $M_j = M(Z)/k$. The gap $\Delta = k\log(M(Z)/k) - \sum_j \log M_j \;\geq 0$ is invariant under a global rescaling $M \mapsto \kappa M$, and $1 - e^{-\Delta}$ is the equivalent proportional shortfall in the Nash product.

Proof. The inequality is Proposition 3 with no structural constraint imposed, i.e. the case $C = 1$ of the component bound of Section 9: the total $M(Z)$ is partition-invariant, and $\sum_j \log m_j$ is maximised over $\{m > 0 : \sum_j m_j = M(Z)\}$ uniquely at $m_j = M(Z)/k$. For invariance, rescaling sends $\sum_j \log M_j$ to $\sum_j \log M_j + k \log \kappa$ and $k\log(M(Z)/k)$ to $k\log(M(Z)/k) + k\log\kappa$, so Δ is unchanged; the last claim is the definition of Δ as a log-ratio of products. □

This is a free dual bound: it needs $M(Z)$ and k and nothing else — no geometry, no contiguity, no solver — and it holds for *every* partition of these zips into k districts, so a draw's gap to it is an unconditional statement about how much balance was left on the table. What it does not say is whether the ceiling is reachable. Zips are indivisible, so in general it is not, which is the next certificate's subject.

8.2 (ii) The integer balance floor: an honest bound pair

Write $\tau = M(Z)/k$ for the target. The best balance indivisible zips permit, with geometry ignored entirely, is the optimum t^* of

$$\begin{aligned} \min_{x,t} \quad & t \\ \text{s.t.} \quad & \sum_j x_{zj} = 1 \qquad && \forall z \in Z, \\ & \Big|\sum_z M_z x_{zj} - \tau\Big| \leq t && \forall j, \\ & x_{zj} \in \{0,1\}, \quad t \geq 0. \end{aligned} \tag{12}$$

Every real draw's max-deviation is at least t^*, so t^* separates the heuristic's loss from the arithmetic's: it says how much of the gap to Proposition 9's ceiling was ever available. The trouble is that (12) is a multiway-number-partitioning problem, and its relaxation is worthless.

Proposition 10 (The relaxation of (12) is vacuous)**.** *The linear relaxation of (12), replacing $x_{zj} \in \{0,1\}$ by $x_{zj} \geq 0$, has optimal value 0 for every instance and every $k \geq 1$.*

Proof. Take $x_{zj} = 1/k$ for all z, j. The placement rows hold, and $\sum_z M_z x_{zj} = M(Z)/k = \tau$ for every j, so $t = 0$ is feasible; $t \geq 0$ is imposed, so 0 is optimal. □

The root bound therefore carries no information, and every nat of the dual side must be earned in the tree — against the full $k!$ label symmetry, which is what branch-and-bound cannot prune. Two valid symmetry breaks help and do not fix it: fix the heaviest zip into district 0 (a relabelling, so it costs nothing), and force districts $1, \ldots, k-1$ to non-increasing mass (valid because after fixing that one zip those labels remain freely permutable).

So the certificate is a **pair**, and it is reported as one. The lower bound is 0 from Proposition 10 — or whatever the solver has earned above it within its time limit. The upper bound is *constructive*: any partition anyone exhibits is an upper bound on t^*, needing no proof beyond arithmetic, so one is built directly by longest-processing-time greedy (heaviest zip first, into the lightest district) followed by a steepest single-move/single-swap descent on the max-deviation. Both take a fraction of a second, the partition is returned with the certificate, and its masses are recomputed from it. The primal half is the operative one for judging a draw: it is what says whether the draw's imbalance is arithmetic or geometry.

8.3 (iii) The pinned-centers assignment MILP

The third certificate asks the geometric question with the draw's own centers held fixed: minimise $\sum_{z,j} M_z \|x_z - c_j\|^2 y_{zj}$ subject to $\sum_j y_{zj} = 1$, $|\sum_z M_z y_{zj} - \tau| \leq \delta$ and y binary, with δ defaulting to the draw's own max-deviation so that the draw is feasible for its own test. Pinning the centers removes the label symmetry entirely, and what remains is the transportation problem of Lemma 7 with two side rows per district — so the relaxation is nearly integral, the root bound is tight, and a real certificate (`mip_rel_gap` = 0, trap 12) closes in minutes at production size.

Remark 2 (What (iii) does not certify)**.** The centers are the heuristic's. Certificate (iii) proves the *assignment* is optimal *given* them, in exactly the sense a k-means assignment step is optimal given its centroids; it says nothing about whether those k points are the right ones, and the joint problem over centers *and* assignment is untouched by anything here. Nor does it certify the stage-1 objective: the constraint is a max-deviation band, not $\sum_j \log M_j$, so a strictly more compact assignment inside the band may have slightly *lower* Nash value. Both values are returned so the trade is visible rather than implied.

8.4 The real run

Numbers below are from run `battery/results/draw_k13_20260901` (the portfolio) and the certification run of 2026-09-01 (the three certificates on that portfolio's winner), both on the descaled export at $k = 13$ over the 1,223 zips carrying coordinates. Masses are in descaled units; every quoted spread, ratio and nat is scale-free by Proposition 2.

The portfolio did its job. Five seeds were drawn and scored by stage 2. The winner is seed 3, at a staffing value of 59.9375 nats, and it is *not* the stage-1-best draw: its stage-1 Nash value is 69.563100 against the best draw's 69.563143, a stage-1 shortfall of 4.38×10^{-5} nats bought for 7.12×10^{-2} nats of staffing value. That is the split-stage mitigation of Section 6.2 working exactly as designed — and a reminder that stage-1 optimality is not the quantity of interest.

The draw. The winner's 13 districts span 0.642% of the target (max-deviation 0.4005%); after the 6 coordinate-less zips are placed by the state-plurality rule the spread is 0.781%. Its final transportation LP split exactly $12 = k - 1$ zips, the bound of Lemma 7(3).

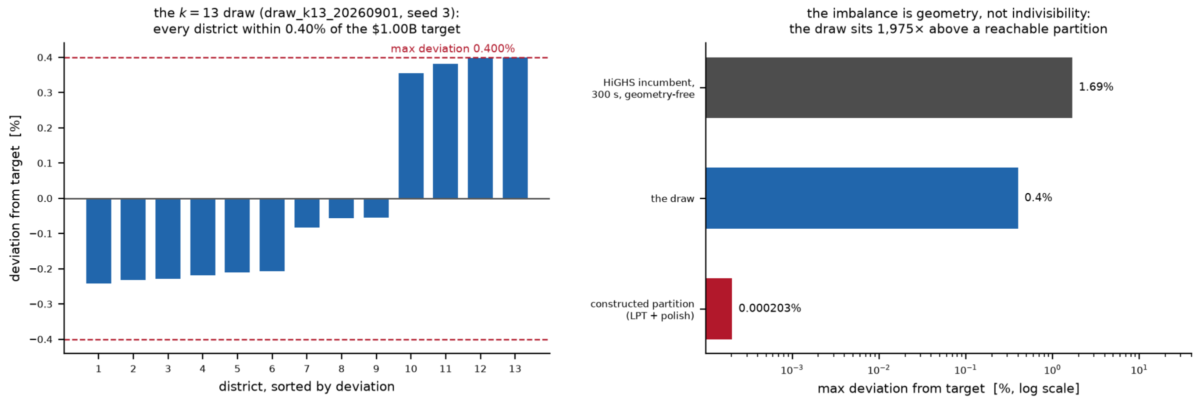

Figure 1: The $k = 13$ draw of `draw_k13_20260901` and what the certificates say about it. *Left:* each district's deviation from the $1.00B target; the dashed lines are the realised max-deviation of 0.4005%. *Right, log scale:* the same max-deviation against the two reference points of certificate (ii) — a constructed partition proving 0.00020% is reachable when geometry is ignored, and the geometry-free MILP's own 300 s incumbent. The draw sits three orders of magnitude above the reachable floor and an order of magnitude below the MILP's incumbent: its imbalance is bought by compactness, not forced by indivisible zips.

The certificates.

- **Ceiling (proved, analytic).** The draw scores 69.563100 nats against a ceiling of 69.563145: a gap of 4.510×10^{-5} nats, or 0.0045% of the Nash product. *Proved:* no partition of these zips into 13 districts scores above the ceiling, whatever its geometry. *Not proved:* that the ceiling is reachable.

- **Integer balance floor (bound pair).** $t^* \in [0,\ 2.028 \times 10^{-6}]$ as a fraction of target. The upper end is constructive — an explicit partition from the greedy plus polish — and the lower end is Proposition 10's vacuous zero: with t capped at the constructed value the MILP explored 0 nodes in 300 s and returned no incumbent, and uncapped its own best after 300 s was a max-deviation of 1.69%, worse than the geometric draw it was meant to bound. The reading does not depend on closing that gap: the draw sits at 0.4005%, about 1,975 times a *demonstrably reachable* balance, so **the draw's residual imbalance is the price of geometry, not of indivisibility**.

- **Pinned-centers assignment (proved).** In 168 s and 62,659 nodes, with the gap closed to zero, the MILP proved that an assignment 8.53% more compact than the draw exists at the same balance band, relabelling 152 zips at a cost of -4.66×10^{-5} nats of stage-1 Nash. It has *not* been adopted: which of balance and compactness is the tie-break for the other is a decision, not a solver setting, and it is recorded as an open question in Section 11. Per Remark 2 this says nothing about the centers.

9 The balance ceiling under a disconnected footprint (superseded as a planning tool)

Why this section is demoted. It was written when the footprint was believed to be four disconnected regions — two coasts, Texas, Florida — and its purpose was to decide k before any solver ran, from four regional totals. The export refuted the premise: the footprint is national (Section 1), and under center-based stage 1 the map is one connected canvas, so there are no components to allocate districts across. The propositions below are unaffected — they are statements about partitions of a disconnected set, and Proposition 11 at $C = 1$ is exactly certificate (i) of Section 8. What is withdrawn is the *illustrative table* that followed them: computed on a \$6.2B total across four components, it reported ceiling spreads from 19.4% to 87.1% at $k = 6$ and concluded that \$1B $\pm$ 10% was probably geometrically unreachable. Both inputs were wrong, and the real answer runs the other way: the drawn map holds every district within 0.4005% of target. The section is kept because the argument becomes live again the moment a genuinely disconnected sub-problem appears — a single-state carve-out, say.

9.1 The bound

Let G have connected components $K_1, \dots, K_C$ with masses $M_c = \sum_{z \in K_c} M_z > 0$. A contiguous district is connected, hence contained in a single component. So a partition of Z into k contiguous districts induces a *district budget* $(k_1, \dots, k_C)$ with $k_c \geq 1$ and $\sum_c k_c = k$; feasibility requires $k \geq C$.

Proposition 11 (Balance ceiling). *Every partition of Z into k contiguous districts satisfies*

$$\sum_{j=1}^{k} \log M_j \quad \leq \quad \mathrm{UB}(k) \ := \max_{k_c \geq 1,\ \sum_c k_c = k} \ \sum_{c=1}^{C} k_c \log \frac{M_c}{k_c}, \tag{13}$$

and $\mathrm{UB}(k) = -\infty$ *(no feasible partition exists) when* $k < C$.

Proof. Fix a contiguous k-partition and let (k_c) be the budget it induces; $k_c \geq 1$ because K_c is nonempty and must be covered, and $\sum_c k_c = k$ because each district lies in exactly one component. The districts inside K_c partition K_c, so by Proposition 3 applied within the component, $\sum_{j \subseteq K_c} \log M_j \leq k_c \log(M_c/k_c)$. Summing over c bounds the partition's objective by $\sum_c k_c \log(M_c/k_c)$, which is at most the maximum over budgets. If $k < C$ then some component receives no district and is uncovered, so no partition exists. $\square$

At $C = 1$ the maximisation is vacuous and (13) reads $\sum_j \log M_j \leq k \log(M(Z)/k)$ — Proposition 9, the certificate actually used. The implementation (`channel.allocate_districts`) enumerates budgets directly, and `test_ceiling_is_an_upper_bound_on_any_real_partition` checks the inequality against brute force on every contiguous three-way cut of a path.

Corollary 12 (Small components force undersized districts). *If some component has $M_c < \tau$ for a target territory size τ, then every contiguous partition contains a district of mass at most $M_c < \tau$. In particular a $\pm\epsilon$ band around τ is infeasible whenever $M_c < (1 - \epsilon)\tau$, for every k and every algorithm.*

Proof. The districts inside K_c partition it, so the smallest has mass at most $M_c/k_c \leq M_c$. $\square$

Proposition 13 (Spread floor)**.** *Fix a district budget* (k_c)*. Every contiguous partition realising it has*

$$\frac{\max_j M_j - \min_j M_j}{M(Z)/k} \;\geq\; \frac{\max_c M_c/k_c - \min_c M_c/k_c}{M(Z)/k},$$

the right-hand side being the spread of the ceiling for that budget. Minimising over budgets gives an unconditional lower bound on the spread of any contiguous k*-partition.*

Proof. Inside component c the k_c district masses average M_c/k_c, so the largest is $\geq M_c/k_c$ and the smallest is $\leq M_c/k_c$. Taking the maximum of the former over c and the minimum of the latter over c gives $\max_j M_j \geq \max_c M_c/k_c$ and $\min_j M_j \leq \min_c M_c/k_c$; the mean $M(Z)/k$ is fixed. The unconditional statement follows by minimising over the finitely many budgets. □

Remark 3 (The two optima need not coincide)**.** The budget maximising the objective (13) is not always the budget minimising the ceiling spread, so the two bounds must be reported from different budgets and never quoted interchangeably. In the withdrawn illustrative table this happened in exactly one cell: the east-heavy split's objective-optimal budget gave a ceiling spread of 87.1% at $k = 6$, while the spread-minimising budget reached 77.4% at a lower objective. Both are valid bounds of their respective kinds; neither dominates. `channel.allocate_districts` returns both (`ceiling_spread_rel` and `min_spread_rel`, with `spread_optima_agree`).

10 The contiguity formulation (superseded by the data)

Why this section is demoted. This was the stage-1 formulation until the export arrived. It fell for one empirical reason: the sold-zip Rook adjacency graph has 547 components, the largest holding 5.1% of opportunity and 68% of opportunity sitting in components below 1% each, so a contiguity constraint at $k = 13$ is not hard, it is infeasible — and on any graph where it *is* feasible over sold zips it is nearly vacuous, since almost every district would be a crumb. Contiguity was a proxy for coverability, and Section 7 optimises that requirement directly instead. The formulation is kept intact because its four notes are traps already paid for in the two-player programme, and any future contiguity variant — a re-export including the unsold glue zips, a single-state carve-out, a county-level aggregation — will meet all four again.

Written in the n-way form the harness carries, with $y_{z,i} = 1$ iff zip z is assigned to candidate i:

$$\begin{aligned}
\max \quad & \sum_i \zeta_i \;-\; \rho \sum_{e \in E} w_e \\
\text{s.t.} \quad & \sum_{i \in \mathrm{cand}(z)} y_{z,i} = 1 && \forall z \in Z, \\
& g_i \;\leq\; \sum_z u_i(z)\, y_{z,i} && \forall i, \\
& \zeta_i \;\leq\; \log \hat{g}_i + \frac{\left(\sum_z u_i(z) y_{z,i}\right) - \hat{g}_i}{\hat{g}_i} && \forall i,\ \forall \text{ tangent points } \hat{g}_i > 0, \\
& \sum_{w \in C} y_{w,i} \;\geq\; y_{u,i} + y_{v,i} - 1 && \forall i,\ \forall u, v,\ C \text{ a } u,v\text{-separator}, \\
& y_{z,i} \in \{0,1\}.
\end{aligned} \tag{14}$$

- **$\leq$, not $=$, on the gains.** The objective is increasing in g_i through the logs, so $g_i \leq \sum_z u_i(z) y_{z,i}$ is tight at every optimum. Writing it as an equality lets presolve multi-aggregate the gain variables out of the transformed problem, after which no in-callback incumbent can be posted at all. Relatedly, dual presolve reductions must be disabled: they count variable locks over the constraints presolve can *see*, and lazily separated rows are not there yet.

- **The tangents are outer approximation, one family per representative.** Each supports $\zeta_i \leq \log g_i$ at $\hat{g}_i$ and is valid everywhere, so the tangent set is a relaxation and the master's value is a genuine upper bound (Duran and Grossmann, 1986; Fletcher and Leyffer, 1994); a certificate is the moment the incumbent meets it. The ε-certified piecewise-linear alternative (Vielma and Nemhauser, 2011; Caragiannis et al., 2019) trades the loop for a bounded a-priori error. The gain lower bound must come from the incumbent, never from an absolute floor like 10^{-9}: the log's gradient there is 10^9 and destabilises the LP.

- **The separator cuts are root-free and uniform.** If a feasible allocation puts both u and v with representative i, it contains an i-path between them; every such path meets the separator C, so some $w \in C$ has $y_{w,i} = 1$. If either endpoint is elsewhere the right-hand side is ≤ 0 and the cut is slack. Validity does not depend on how u, v, C were chosen, which is what lets the same construction separate fractional points. Crucially, *no root is fixed*: a rooted formulation is unsound on a disconnected graph, since it forces growth toward a vertex other components cannot reach, and its "dual bound" can sit strictly below feasible allocations already visited — and this footprint's graph is disconnected 547 ways. Encoding alternatives — single-commodity flow from a root (Shirabe, 2005), lazily separated cuts (Validi et al., 2021; Validi and Buchanan, 2022), cutting-plane aggregation (Oehrlein and Haunert, 2017) — differ mainly in how they pay for this.

- **Both cut families belong in *one* branch-and-cut tree.** Outer approximation converges by the Duran–Grossmann argument and lazy feasibility cuts by combinatorial Benders (Hooker and Ottosson, 2003; Codato and Fischetti, 2006); in a multi-tree loop that restarts a fresh MILP after each round, the two iteration counts *multiply*. A single tree with both families separated lazily at nodes (Quesada and Grossmann, 1992; Bonami et al., 2008; Validi et al., 2021) keeps one globally valid dual bound throughout.

The symmetry hazard, and the Hess remedy. Formulation (14) inherited its index i from the merger problem, where representatives are named and distinguishable. In stage 1 they are not: the k districts are *anonymous*, so any permutation of the labels maps a feasible solution to an equally good one and branch-and-bound must re-derive and re-prune $k!$ copies of every subtree. The standard remedy is the centre-based formulation of Hess and Samuels (1971): index districts by a centre and pin the labels by geography (Zoltners and Sinha, 2005; Salazar–Aguilar et al., 2011; Ríos-Mercado and Pérez, 2012). Section 7 is that remedy, adopted for a second reason as well — and Proposition 10 is the same hazard surviving into the one place the certificates still meet it, the geometry-free balance floor.

If an exact contiguous model is ever needed at this size. In increasing order of concession: per-component solves; pre-aggregation to clusters or county/CBSA level, then disaggregation (Karypis and Kumar, 1998); ε-certified acceptance at the harness's existing tier-2 tolerance; and last a heuristic with a reported bound, for which Proposition 11 supplies the dual side for free and column generation or recombination sampling the primal (Gurnee and Shmoys, 2021; DeFord et al., 2021; Bozkaya et al., 2003). Connected fair division is NP-hard already for two agents with identical valuations (Deligkas et al., 2021), so exactness with certificates — not a closed form — is the realistic target; see Bouveret et al. (2017); Bilò et al. (2022); Suksompong (2019) for the fair-division-on-graphs setting and Bei et al. (2022) for the only price-of-connectivity bounds available (none is known for Nash welfare).

11 Open questions

These are decisions, not implementation gaps. Each changes the answer.

1. **Lexicographic priority: balance or compactness?** Certificate (iii) proved that an assignment 8.53% more compact than the drawn map exists inside the same max-deviation band, relabelling 152 zips for -4.66×10^{-5} nats of stage-1 Nash. Adopting it means declaring balance-within-a-band the constraint and compactness the objective; declining it means the reverse. The current pipeline declines by default, which is a choice made silently and should be made explicitly — along with the band's width, the one number that makes the question well-posed.

2. **Center choice.** Every certificate here is conditional on the centers, which come from k-means++ seeding and Lloyd rounds. The exact question — choose the k centers *and* the assignment jointly, as Hess's own formulation does with centre binaries $y_{c,c}$ and $y_{z,c} \leq y_{c,c}$ — is untouched, and at $1{,}223 \times 13$ it is the natural next MILP. Until it is solved, the honest statement is the one Remark 2 makes: the assignment is certified, the districts are not.

3. **Empty bundles.** Insisting that every representative receive a non-empty bundle is a real and possibly infeasible constraint, and a single $g_i = 0$ sends $\sum_i \log g_i$ to $-\infty$. The standard treatment is lexicographic — maximise the count of representatives with positive utility, then maximise Nash welfare among them (Caragiannis et al., 2019) — and is the recommendation. Under rectangular stage-2 matching the question is currently moot (111 representatives, 13 districts, none unstaffed), but it returns the moment k approaches the headcount.

4. **Directionality of θ.** If capture depends on *which* representative is displaced, the single scalar becomes $\theta_{i \leftarrow j}$ and the identification problem grows from one parameter to $O(n^2)$ — and θ is already the one parameter resting on unfinished causal work.

5. **The filler capture mode.** $c_{\text{free}} \in \{c_2, c_1, \lambda\}$ per Section 2.3; `full` (c_1) is the recommendation, `theta` is the default only because it is the no-change case. At 2 vacant zips it cannot move this map, so the decision is cheap to take and should be taken before it is not.

6. **The brute-force acceptance tier.** The harness accepts a method partly on matching exhaustive enumeration at $n \leq 20$; at three candidates per zip that is $3^{20} \approx 3.5 \times 10^9$ leaves. The tier must be re-cut by candidate-weighted size $\prod_z |\text{cand}(z)| \leq 10^6$–$10^7$. This weakens a stated acceptance criterion and needs sign-off, not a quiet redefinition.

7. **The staffing ratio.** 111 representatives against 13 territories is 8.5:1. Stage 2's rectangular matching reads the unmatched as *not selected for this channel*; if instead the channel is meant to absorb the whole force, k is wrong and the target is not \$1B.

References

Xiaohui Bei, Ayumi Igarashi, Xinhang Lu, and Warut Suksompong. The price of connectivity in fair division. *SIAM Journal on Discrete Mathematics*, 2022. doi: 10.1137/20m1388310.

Vittorio Bilò, Ioannis Caragiannis, Michele Flammini, Ayumi Igarashi, Gianpiero Monaco, Dominik Peters, Cosimo Vinci, and William S. Zwicker. Almost envy-free allocations with connected bundles. *Games and Economic Behavior*, 2022. doi: 10.1016/j.geb.2021.11.006.

Pierre Bonami, Lorenz T. Biegler, Andrew R. Conn, Gérard Cornuéjols, Ignacio E. Grossmann, Carl D. Laird, Jon Lee, Andrea Lodi, François Margot, Nicolas Sawaya, and Andreas Wächter. An algorithmic framework for convex mixed integer nonlinear programs. *Discrete Optimization*, 2008. doi: 10.1016/j.disopt.2006.10.011.

Sylvain Bouveret, Katarína Cechlárová, Edith Elkind, Ayumi Igarashi, and Dominik Peters. Fair division of a graph. In *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence (IJCAI 2017)*, 2017. doi: 10.24963/ijcai.2017/20.

Burcin Bozkaya, Erhan Erkut, and Gilbert Laporte. A tabu search heuristic and adaptive memory procedure for political districting. *European Journal of Operational Research*, 2003. doi: 10.1016/s0377-2217(01)00380-0.

Ioannis Caragiannis, David Kurokawa, Hervé Moulin, Ariel D. Procaccia, Nisarg Shah, and Junxing Wang. The unreasonable fairness of maximum Nash welfare. *ACM Transactions on Economics and Computation*, 2019. doi: 10.1145/3355902.

Gianni Codato and Matteo Fischetti. Combinatorial Benders' cuts for mixed-integer linear programming. *Operations Research*, 2006. doi: 10.1287/opre.1060.0286.

Daryl DeFord, Moon Duchin, and Justin Solomon. Recombination: A family of Markov chains for redistricting. *Harvard Data Science Review*, 2021. doi: 10.1162/99608f92.eb30390f.

Argyrios Deligkas, Eduard Eiben, Robert Ganian, Thekla Hamm, and Sebastian Ordyniak. The parameterized complexity of connected fair division. In *Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence (IJCAI 2021)*, 2021. doi: 10.24963/ijcai.2021/20.

Juan C. Duque, Richard L. Church, and Richard S. Middleton. The p-regions problem. *Geographical Analysis*, 2011. doi: 10.1111/j.1538-4632.2010.00810.x.

Marco A. Duran and Ignacio E. Grossmann. An outer-approximation algorithm for a class of mixed-integer nonlinear programs. *Mathematical Programming*, 1986. doi: 10.1007/BF02592064.

E. Eisenberg and David Gale. Consensus of subjective probabilities: The Pari-Mutuel method. *The Annals of Mathematical Statistics*, 1959. doi: 10.1214/aoms/1177706369.

Roger Fletcher and Sven Leyffer. Solving mixed integer nonlinear programs by outer approximation. *Mathematical Programming*, 1994. doi: 10.1007/BF01581153.

Wes Gurnee and David B. Shmoys. Fairmandering: A column generation heuristic for fairness-optimized political districting. In *SIAM Conference on Applied and Computational Discrete Algorithms (ACDA21)*, 2021. doi: 10.1137/1.9781611976830.9.

Sidney W. Hess and Stuart A. Samuels. Experiences with a Sales Districting Model: Criteria and implementation. *Management Science*, 1971. doi: 10.1287/mnsc.18.4.p41.

J. N. Hooker and G. Ottosson. Logic-based Benders decomposition. *Mathematical Programming*, 2003. doi: 10.1007/s10107-003-0375-9.

Ehud Kalai and Meir Smorodinsky. Other solutions to Nash's bargaining problem. *Econometrica*, 1975. doi: 10.2307/1914280.

George Karypis and Vipin Kumar. Multilevel k-way partitioning scheme for irregular graphs. *Journal of Parallel and Distributed Computing*, 1998. doi: 10.1006/jpdc.1997.1404.

John F. Nash. The bargaining problem. *Econometrica*, 1950. doi: 10.2307/1907266.

Johannes Oehrlein and Jan-Henrik Haunert. A cutting-plane method for contiguity-constrained spatial aggregation. *Journal of Spatial Information Science*, 2017. doi: 10.5311/josis.2017.15.379.

I. Quesada and I. E. Grossmann. An LP/NLP based branch and bound algorithm for convex MINLP optimization problems. *Computers & Chemical Engineering*, 1992. doi: 10.1016/0098-1354(92)80028-8.

Roger Z. Ríos-Mercado and Elena Fernández. A reactive GRASP for a commercial territory design problem with multiple balancing requirements. *Computers & Operations Research*, 2009. doi: 10.1016/j.cor.2007.10.024.

Roger Z. Ríos-Mercado and López Pérez. Commercial territory design planning with realignment and disjoint assignment requirements. *Omega*, 2012. doi: 10.1016/j.omega.2012.08.002.

M. Angélica Salazar–Aguilar, Roger Z. Ríos-Mercado, and Mauricio Cabrera-Ríos. New models for commercial territory design. *Networks and Spatial Economics*, 2011. doi: 10.1007/s11067-010-9151-6.

Takeshi Shirabe. A model of contiguity for spatial unit allocation. *Geographical Analysis*, 2005. doi: 10.1111/j.1538-4632.2005.00605.x.

Warut Suksompong. Fairly allocating contiguous blocks of indivisible items. *Discrete Applied Mathematics*, 2019. doi: 10.1016/j.dam.2019.01.036.

Hamidreza Validi and Austin Buchanan. Political districting to minimize cut edges. *Mathematical Programming Computation*, 2022. doi: 10.1007/s12532-022-00221-5.

Hamidreza Validi, Austin Buchanan, and Eugene Lykhovyd. Imposing contiguity constraints in political districting models. *Operations Research*, 2021. doi: 10.1287/opre.2021.2141.

Juan Pablo Vielma and George L. Nemhauser. Modeling disjunctive constraints with a logarithmic number of binary variables and constraints. *Mathematical Programming*, 2011. doi: 10.1007/s10107-009-0295-4.

Andris A. Zoltners and Prabhakant Sinha. The 2004 ISMS practice prize winner—sales territory design: Thirty years of modeling and implementation. *Marketing Science*, 2005. doi: 10.1287/mksc.1050.0133.